

These models already produced decent results, but our hopes for Tree-based models were rightfully even higher, as Random Forest was the model of choice in the original study which published the previous version of this dataset.

In our updated dataset XGBoost achieved the best performance with accuracy over 96% and a ROC-AUC of 0.98. Looking at the confusion matrices of the tree based Methods, we can see, that XGBoost only produced 2 false negatives and 4 false positives. These results are promising, as they go towards a direction where these models might be clinically applicable.

If we take a brief look at feature importance Appendix_Diameter and Ipsilateral_Rebound_Tenderness were variables that were important for all TBM. The boxplot here shows, that having a bigger diameter of the appendix on the US is highly correlated with a positive diagnosis for appendicitis. While a model without indicator variables and the unaltered dataset (with NaNs) still performed similar, one could argue that in order to avoid any verification bias you would need to throw out certain variables which might not be available at the moment a patient is still „new“ to the hospital, which limits the use for first diagnostic assessment.

Related work that used more complex models fortunately achieved even higher accuracy with crossattentional mechanisms and image analysis. They also build a web application that is prepared to be used soon, as we hope ML can improve medical diagnostics and can save lives. We still think our contribution is meaningful, as it represents a benchmark for more sophisticated models to beat and is applicable in various data settings.

Here are our sources, as the original study and the following improvement attempts are worthwhile reading.